

Setup Guide

The guide helps you setup Joule Desktop, Claude Code, Connect to HANA backend, Create and Maintain Skills used in this workshop.

Joule Desktop:

Install Joule Desktop using Self Service App

Setup HANA MCP Server

Prerequisites

Make sure you have Node.js 18+ installed. Check by running the following command in your terminal:

```
node --version
```

```
npm --version
```

Step 1 — Install hana-mcp-server

Open a new terminal and enter the following command:

```
npm install -g hana-mcp-server
```

Verify it is installed correctly by running the command:

```
hana-mcp-server --version
```

Step 2 — Set Environment Variables

In the terminal and enter the following command:

```
vi ~/.zshrc
```

Add the following to list of environment variables:

```
export HANA_HOST=ecf4486c-c6ab-4def-bd2b-0ea86311835e.hna1.prod-us10.hanacloud.ondemand.com
```

```
export HANA_PORT=443
```

```
export HANA_USER=DBADMIN
```

```
export HANA_PASSWORD=Institute2026
```

```
export HANA_SCHEMA=DBADMIN
```

```
export HANA_ENCRYPT=true
```

```
export HANA_VALIDATE_CERT=false
```

```
export HANA_ALLOW_INSERT=true
export HANA_ALLOW_UPDATE=true
export HANA_ALLOW_DELETE=true
```

Press Escape and then enter :wq to save and exit and then run the following command:

```
source ~/.zshrc
```

Step 3 — Bridge stdio → HTTP (required for Joule Work Desktop)

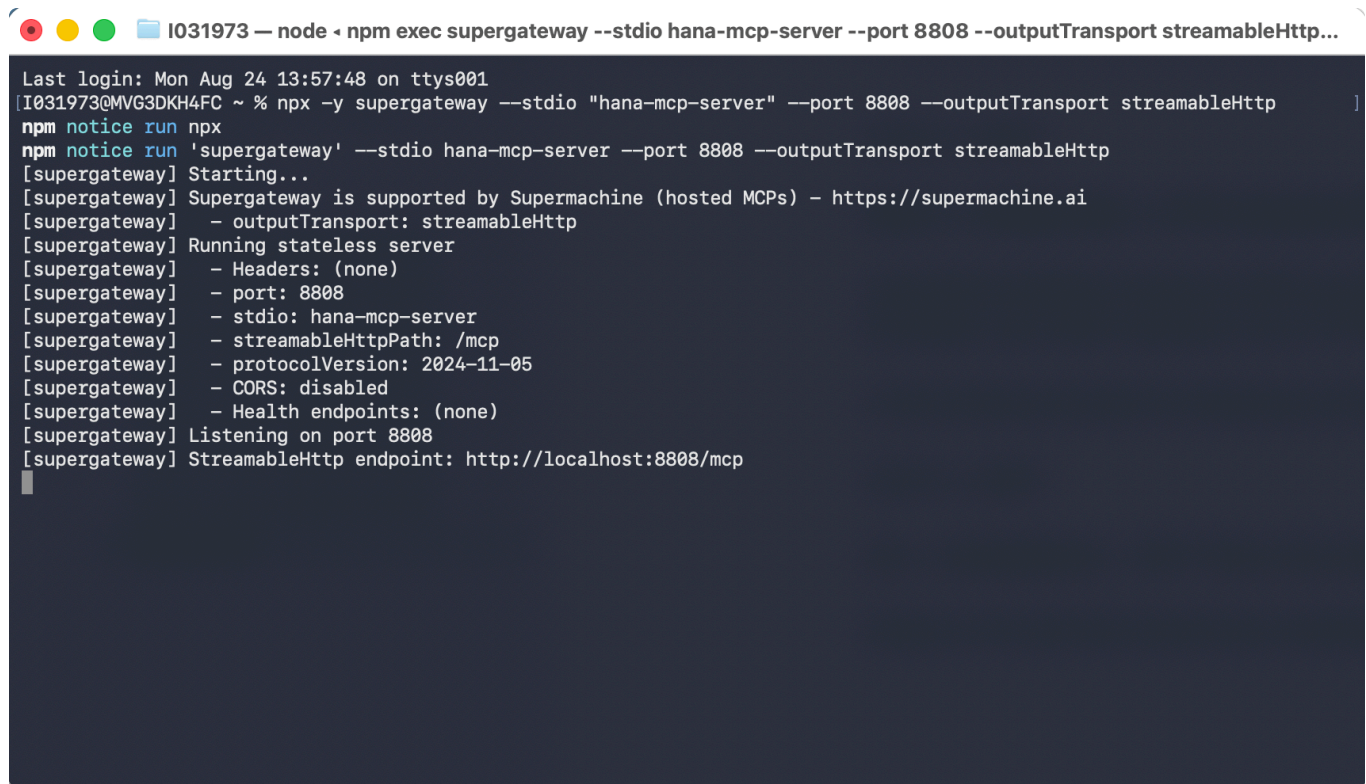
Joule Work Desktop connects to MCP servers via a URL. Since hana-mcp-server runs over stdio by default, you need to expose it as an HTTP/SSE endpoint using **supergateway**:

In the terminal and enter the following command:

```
source ~/.zshrc
```

```
npx -y supergateway --stdio "hana-mcp-server" --port 8808 --outputTransport streamableHttp
```

Output on the terminal should look something like this:



```
I031973 — node • npm exec supergateway --stdio hana-mcp-server --port 8808 --outputTransport streamableHttp...

Last login: Mon Aug 24 13:57:48 on ttys001
I031973@MVG3DKH4FC ~ % npx -y supergateway --stdio "hana-mcp-server" --port 8808 --outputTransport streamableHttp
npm notice run npx
npm notice run 'supergateway' --stdio hana-mcp-server --port 8808 --outputTransport streamableHttp
[supergateway] Starting...
[supergateway] Supergateway is supported by Supermachine (hosted MCPs) - https://supermachine.ai
[supergateway] - outputTransport: streamableHttp
[supergateway] Running stateless server
[supergateway] - Headers: (none)
[supergateway] - port: 8808
[supergateway] - stdio: hana-mcp-server
[supergateway] - streamableHttpPath: /mcp
[supergateway] - protocolVersion: 2024-11-05
[supergateway] - CORS: disabled
[supergateway] - Health endpoints: (none)
[supergateway] Listening on port 8808
[supergateway] StreamableHttp endpoint: http://localhost:8808/mcp
```

Step 4 — Add the Connector in Joule Work Desktop

Click the Extensions icon in the left navigation pane

Go to the Connectors tab

Click "+ New" → "Add Connector"

Fill in:

Name: SAP HANA

URL: <http://localhost:8808/mcp>

Click OK

The connector will appear in your list. Joule will automatically discover all the tools it exposes (schema browsing, SQL query, table listing, etc.).

Step 5 — Activate in a Conversation

The connector is not auto-active in every chat. To use it:

Open a new conversation

In the chat input toolbar, click the connector picker

Select SAP HANA

Then ask naturally — for example:

"List all tables in the DBADMIN schema"

"Fetch the first 20 rows from the EMPLOYEE table"

"Describe the structure of the EMPLOYEE table"

This ensures Joule Desktop is now connected to HANA which contains the data required for the workshop.

Claude Code

Step 1: Install Hyperspace LLM Proxy

Click on the following link <https://ai-docs.portal.hyperspace.tools.sap/llm-proxy/quickstart/>

Follow the steps listed and install the Hyperspace LLM Proxy (Graphical or Commandline depending on your comfort)

Start the proxy using **Terminal** or using **Hyperspace AI icon** in your menu bar

Step 2: Install Claude Code

Install and configure Claude Code by following the steps listed in the link below:

<https://ai-docs.portal.hyperspace.tools.sap/llm-proxy/recipes/claude/>

Step 3: Test Claude Code

Open a terminal and run:

